



HS 404: Technology in Pre-modern India

Course Code	HS 404
Course Name	Technology in Pre-modern India
L-T-P-C	3-0-0-3
Intended for	B.Tech.
Distribution	Elective
Approval	5th Senate (Oct 2013)

Course Description

This course introduces students to technology in pre-modern India. It begins by examining the notions of science, which prevailed in India before the advent of the modern period. It then offers an overview of different aspects of science like the mathematics, astronomy, metallurgy, engineering, architecture, agriculture and irrigation. As part of the course-work, students will be made to develop models for sustainable agriculture, irrigation, housing and public utility structures like bridges.

Course Contents

- **Introduction to science in pre-modern India:** i) the nature of science ii) the distinction between science and knowledge, iii) structures of cognition
- **Mathematics and astronomy:** i) Indian systems of mathematics, ii) Advances in astronomical sciences
- **Metallurgy:** i) the advent of copper and bronze, ii) discovery of iron and its implications on Indian history, iii) forms of metallurgy
- **Engineering and architecture:** i) earliest forms of engineering, ii) town planning, ii) housing patterns, iii) religious architecture
- **Agriculture and irrigation:** i) the emergence of agriculture, ii) agrarian expansion and changes in agrarian technology, iii) irrigation

Text Books

1. D M Bose, S N Sen and B V Subbarayappa, A Concise History of Science in India, Indian National Science Academy, New Delhi, 1998.

Readings

1. Debiprasad Chattopadhyaya, History of Science and Technology in Ancient India, Firma K.L. Mukhopadhyaya, Calcutta, 1977.
2. Vibha Tripathi, History of Iron Technology in India, Rupa and Infinity Foundation, New Delhi, 2008.
3. Helaine Selin and RoddamNarasimha, Encyclopedia of Classical Indian Sciences, Universities Press, Hyderabad, 2007.
4. Ashok Jhunjhunwala, Indian Mathematics, Wiley Eastern, New Delhi, 1993.